



CHAPTER

ATA 35

OXYGEN SYSTEM

THIS MANUAL IS INTENDED FOR TRAINING PURPOSE ONLY

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Oxygen - General

Introduction

The oxygen system is part of the emergency equipment installation in the flight compartment and the cabin. In an emergency such as a sudden decrease in cabin pressure or smoke, the oxygen system supplies oxygen for the flight crew and passengers.

General Description

The system supplies oxygen for the crew and the passengers through oxygen tanks, masks, lines and outlets. These are also used to service the system. Portable oxygen and protective breathing equipment is also supplied for the crew to manage in-flight fire, smoke, fumes, and for other passenger emergencies.

The oxygen system includes the subsystems that follow:

- Crew Oxygen System
- Crew Oxygen Delivery
- Crew Oxygen Indication and Servicing
- Passenger Portable Oxygen System
- Passenger Drop Down Oxygen System
- Crew Portable Oxygen
- Crew Portable Oxygen
- Protective Breathing Equipment

Crew Oxygen System

Introduction

The crew oxygen system supplies oxygen to the crew if there is no pressurization, or there is smoke or noxious gases in the flight compartment.

General Description

Refer Figure – Crew Oxygen System

The crew oxygen system has three oxygen masks for the pilot, co-pilot and observer. There is a 39.8 cu ft (1.127 cu m) oxygen cylinder assembly and distribution tubing, along with approved Protective Breathing Equipment (PBE) for fire fighting purposes. Smoke goggles are supplied for crew protection in the flight compartment.

On aircraft with SB84-35-04 or ModSum 4-429556 or ModSum 4-457149 or ModSum 4-457449 or ModSum 4-458062 or ModSum 4-458280 or ModSum 4-458218 incorporated, the crew oxygen system has three full face oxygen masks with smoke goggles for the pilot, copilot and observer.

On aircraft with ModSum 4-459933 incorporated, the crew oxygen system has three full face oxygen masks for the pilot, copilot and observer.

On aircraft with ModSum 4-429541 or ModSum 4-309249 or SB84-35-02 incorporated, a 50.9 cu ft (1.441 cu m) oxygen cylinder assembly with onboard charging facility is installed instead of the 39.8 cu ft (1.127 cu m) oxygen cylinder assembly.

The crew oxygen system can supply the three crew members with supplemental oxygen for a 4 minute descent from 25000 to 14000 feet (7620 to 4267 meters) and 116 minutes at 14000 ft (4267 meters). These calculations are done with the oxygen masks selected to the NORM position.

The system has a lighted gauge on the co-pilot's side console and a second gauge on the oxygen cylinder. The operational pressure of the cylinder is 1850 psi (12755 kPa).

The minimum dispatch pressures for each crew configuration is:

- 1383 psi (9536 kPa) for 2 crew
- 1786 psi (12314 kPa) for 3 crew

On aircraft with ModSum 4-429541 or ModSum 4-309249 or SB84-35-02 incorporated, the minimum dispatch pressures for each crew configuration is:

- 1050 psi (7239.50 kPa) for 2 crew
- 1450 psi (9997.40 kPa) for 3 crew

The system includes the components that follow:

- Oxygen Cylinder Assembly
- Crew Mask and Regulator Assemblies
- Overboard Discharge Indicator
- Outlet Assemblies
- Panel Mounted Pressure Gauge
- Low Pressure Hose Assembly
- High Pressure Hose Assembly
- Capillary Line

The crew oxygen system includes the subsystems that follow:

- Crew Oxygen Delivery
- Crew Oxygen Indication and Servicing
- Protective Breathing Equipment

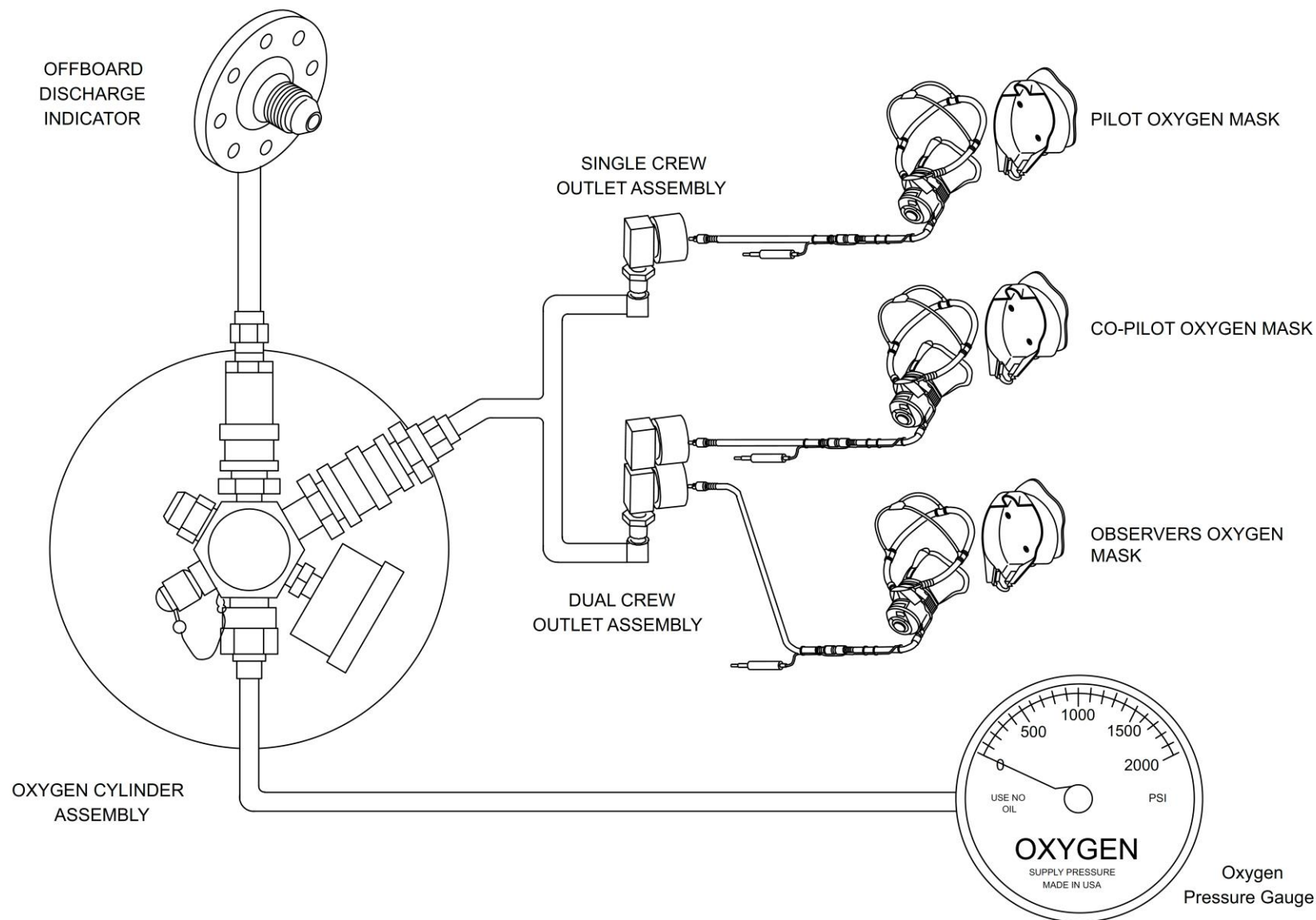


Figure - Crew Oxygen System

Detailed Description

Refer Figure – Crew Oxygen System - Detailed

The oxygen masks are kept in mask stowage cups attached to the bulkhead immediately above and behind the pilot's and co-pilot's seats. The observer's mask is located on the right hand bulkhead.

All three masks are connected to the low pressure distribution lines by individual quick disconnects.

Each mask has an approved regulator with the modes that follow:

- NORM (normal oxygen diluter demand)
- 100% (100% oxygen demand)
- EMER (emergency pressure oxygen).

Each mask has a microphone with a separate audio connector. The masks are of the quick-donning type and will supply oxygen dilution, and if used with the smoke goggles are approved as Protective Breathing Equipment (PBE).

The smoke goggles are kept in pockets in each side console and at the observer's station.

If one mask fails to operate, the observer's mask may be used by either crew member. If either the pilot's or co-pilot's mask fails, the observer is required to use the passenger system.

The crew mask outlets are installed with two outlets on the right side of the flight compartment, and one on the left side. The masks will connect with any crew mask outlet in the flight compartment.

The mask supply hose has an in-line pressure indicator to show if there is sufficient oxygen pressure. The indicator shows green with correct pressure and red if the oxygen pressure is low.

Each hose also has a quick disconnect fitting to connect with the aircraft supply tubing.

The oxygen cylinder is located in the unpressurized lower right nose compartment, rigidly attached by steel clamps to the outside of the nose landing gear bay side wall.

Fabric curtains isolate the cylinder assembly from the hydraulic components in the bay immediately aft, and from the hydraulic tubes below.

The oxygen cylinder must be removed from the aircraft for servicing. The oxygen cylinder pressure gauge shows the cylinder pressure.

The flight compartment gauge shows the amount of oxygen available to the masks. If the oxygen cylinder is turned off, the cylinder low pressure is automatically reduced to atmospheric pressure. If the oxygen cylinder is turned off, the gauge continues to show bottle pressure.

The cylinder is attached to the low pressure distribution tubing by a low pressure hose with a quick disconnect connector. The tubing passes through the pressure bulkhead by a fitting. It then divides and is routed left and right under the flight compartment floor and then up the aft bulkhead to the mask outlets.

The tubing is thin walled stainless steel and all joints are Permaswaged except at the hose and crew mask disconnects which are flareless. The tubing is held in place by cushioned clamps.

High pressure tubing is 0.0625 inch diameter (1.59 millimeter) brass capillary and is routed through the co-pilot's side console to the flight compartment pressure gauge. The connection to the cylinder is by a coil tube.

A high pressure hose connects the safety vent to an overboard discharge indicator located on the right side of the nose.

LEGEND

1. Oxygen cylinder assembly.
2. Pressure regulator.
3. Overboard discharge indicator.
4. Flight compartment oxygen pressure gauge.
5. Co-pilot's oxygen mask.
6. Observer's oxygen mask.
7. Co-pilot and observer oxygen supply outlet.
8. Pilot's oxygen mask.
9. Pilot's oxygen supply outlet.
10. On-board charging valve.

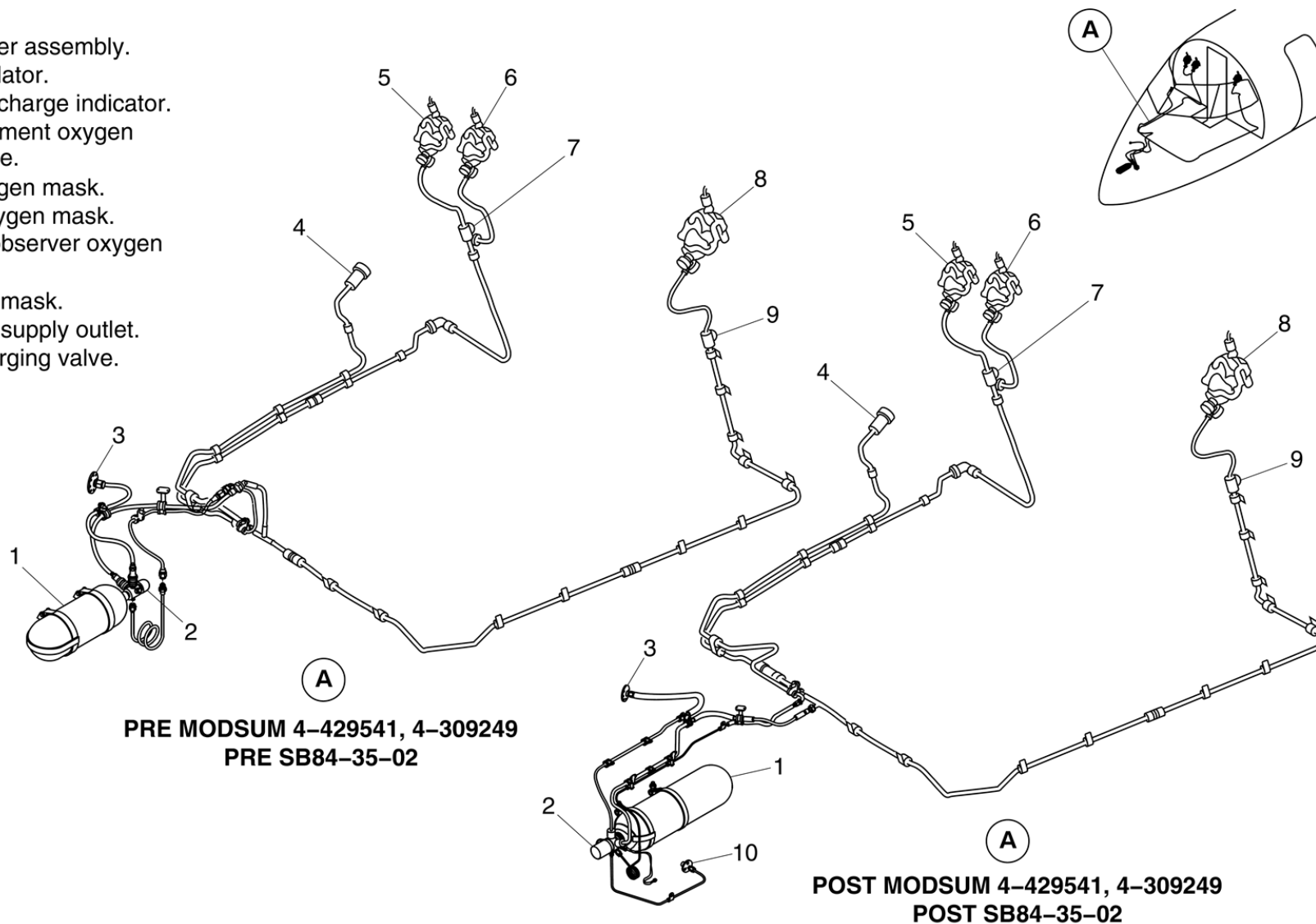


Figure 35-2 Crew Oxygen System – Detailed

Preflight Testing

To test the crew oxygen mask before flight:

- Remove mask from stowage cup by holding the control knob.
- Push the red harness inflation button on the regulator — the harness should inflate smoothly.
- Release the red harness inflation button — the harness should deflate smoothly.
- Put on the mask and breathe normally, check the EMER, 100% and NORM positions — the mask should not vibrate
- the pressure indicator should show green.
- Speak into the mask to check the microphone and volume.
- Take the mask off and put it back in the stowage cup.

To test the crew oxygen mask before flight, without removing it:

- Make sure the audio jack is correctly connected.
- Make sure the oxygen hose is correctly attached to the oxygen outlet.
- Make sure the minimum pressure is supplied by seeing that the pressure indicator shows green.
- Make sure the regulator works by turning the control knob to the EMER position — keeping the control knob in the EMER position can empty the oxygen system.
- The microphone can be checked by listening to the regulator pressurization in the headset.

Emergency Use

To put the mask on in an emergency:

- Remove headset.
- Hold the mask regulator unit with the control knob in the palm of the hand and quickly remove the mask from the cup.
- Hold the sides of the regulator with the thumb and middle finger and push the red inflation button.
- With the hand down and the face seal up, flip the inflated harness out of the way.
- Smoothly lift the inflated mask above the head, with the back of the harness on the back of the neck.
- With the harness hooked behind the head, lower the mask from the forehead to the chin, putting the mask cone on the face.
- Release the red harness inflation button.
- Make sure the mask fits, seals and operates correctly.
- If necessary, replace headset.

Smoke Protection

For smoke protection:

- Put on the crew oxygen mask as above.
- Remove smoke goggles from storage and put them on.
- Tighten the smoke goggle straps.
- Turn the control knob to the EMER position to purge the goggles and prevent infiltration of smoke.
- The control knob can be turned to either the EMER or 100% position as necessary

Leak Test of the Crew Oxygen System

Oxygen system lines must be pressure tested with the high pressure nitrogen test rig. This test makes sure that there are no leaks at all of the fittings or connections between the oxygen system lines and/or between the lines and the functional equipment of the oxygen system.

If there is a leak in the system, this test will find the leak areas. After the necessary repair, the test must be done again until all leaks are eliminated.

Leak test is done in the sequence that follows:

- Pressure test on low pressure lines with nitrogen at 160 psi (1103 kPa) for 15 minutes.
- Purging of low pressure lines with nitrogen at 25 psi (172 kPa) pressure for 30 seconds.
- Pressure test on high pressure lines with nitrogen at 2800 psi (19305 kPa) for 15 minutes.
- Purging of high pressure lines with nitrogen at 250 psi (1724 kPa) pressure for 30 seconds.
- Purging of low pressure lines with oxygen at 70 psi (482 kPa) pressure for 15 seconds.
- Purging of high pressure lines with oxygen by increasing pressure to 1850 psi (12755 kPa) in 10 seconds.
- Final leak test with oxygen.

Purging of lines are done by applying pressure at one end of the line and keep the other ends open. The pressure test and the leak test are done by applying pressure at one end and putting caps on all other ends. Leaks will be detected by comparing the pressures before and after the testing period.

A drop in the final pressure will indicate a leak. Leaks must be located with applicable leak detector solution.

Bonding test is done before introducing oxygen in the lines. This test makes sure that the conductivity between the oxygen carrying components and the aircraft ground is sufficient to prevent hazardous conditions because of the electrical charges.

Crew Portable Oxygen System

Introduction

In addition to the PBE units, the portable oxygen system, dedicated to the flight attendant and passengers, or the flight attendant alone is available as a customer special installation.

General Description

The flight attendant's portable oxygen system has the items that follow:

- a bottle assembly with pressure gauge
- a constant flow outlet
- a low pressure relief valve
- a high pressure frangible safety plug
- an on-off valve
- a charging valve
- a constant flow mask with a flow indicator and a 50 in. (1270 mm) long connecting hose
- a mask reservoir bag and a carrying strap.

The emergency passenger oxygen system has the items that follow:

- a bottle assembly with pressure gauge
- three constant flow outlets with dust plug
- a low pressure relief valve
- a high pressure frangible safety plug
- an on-off valve
- two charging valve
- two constant flow masks with the flow indicators and a 54 in. (1372 mm) long connecting hose
- two reservoir bags for each mask and a carrying strap for oxygen bottle.

Detailed Description

[Refer Figure – Emergency Equipment – General Layout](#)

[Refer Figure – Emergency Equipment – Frwd Draft Bulkhead](#)

[Refer Figure – Emergency Equipment Aft Draft Bulkhead](#)

[Refer Figure – Passenger Emergency Oxygen System](#)

The following is a detailed description of the flight attendant oxygen system and the passenger emergency oxygen system.

Flight Attendant Oxygen System

The flight attendant oxygen system is intended for single person use. It has a mask connected by a hose to the outlet of an oxygen bottle. When required, the oxygen bottle is turned on and the mask is placed over the user's mouth and held in place by an elastic strap. More oxygen flows from the bottle than is needed to allow for leakage.

The flight attendant oxygen bottle is located in a compartment in the G6 galley (Refer SDS 25–30–00). A placard outside of the galley compartment identifies the contents inside. Another placard located inside the compartment identifies the minimum dispatch pressure and the appropriate service manual references for maximum charge pressure and inspection intervals.

Passenger Emergency Oxygen System

The passenger emergency oxygen system is intended for two persons use. It has the masks connected by the hoses to the outlets of an oxygen bottle. When required, the oxygen bottle is turned on and the mask is placed over the user's mouth and held in place by an elastic strap. More oxygen flows from the bottle than is needed to allow for leakage.

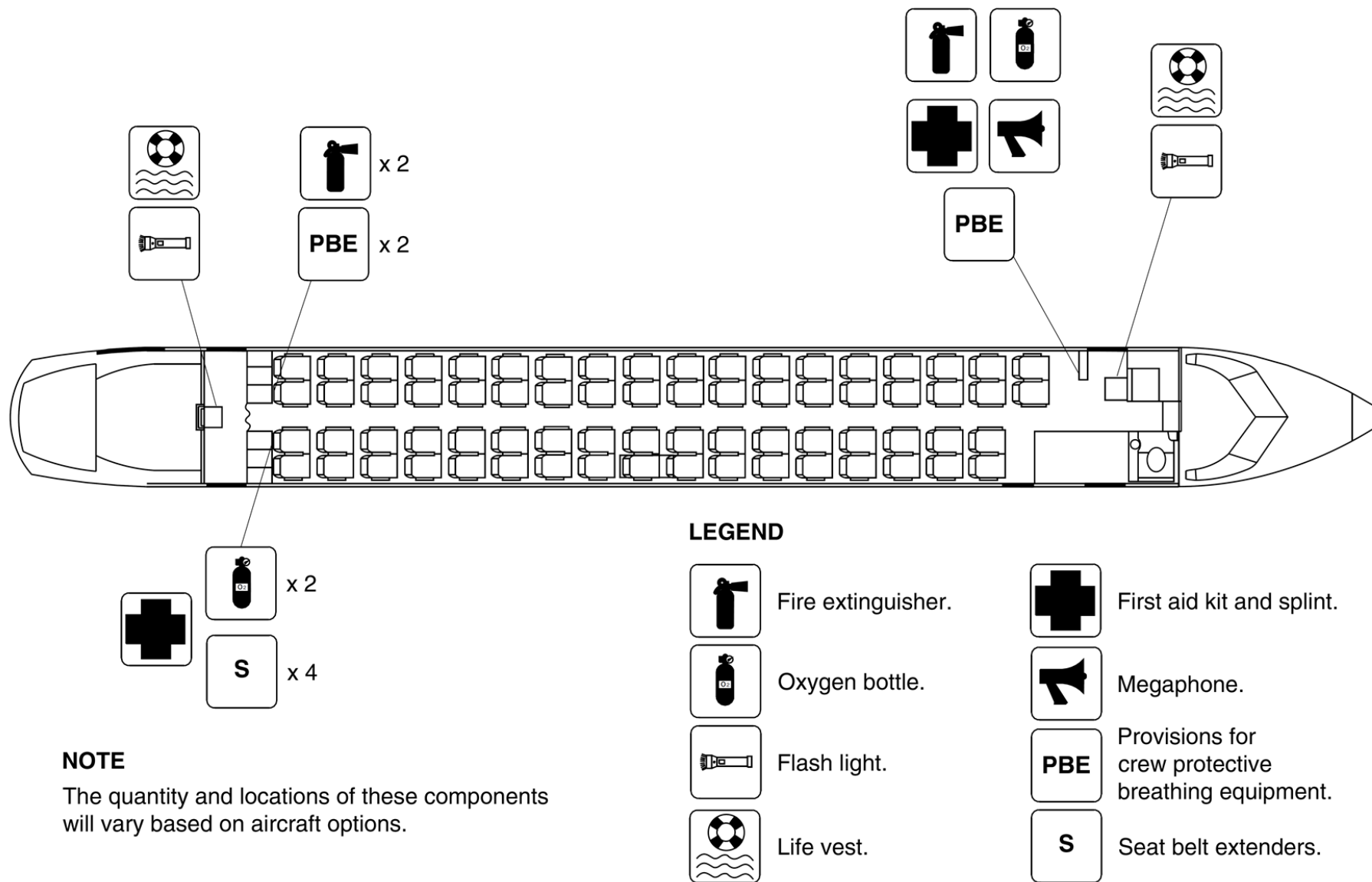
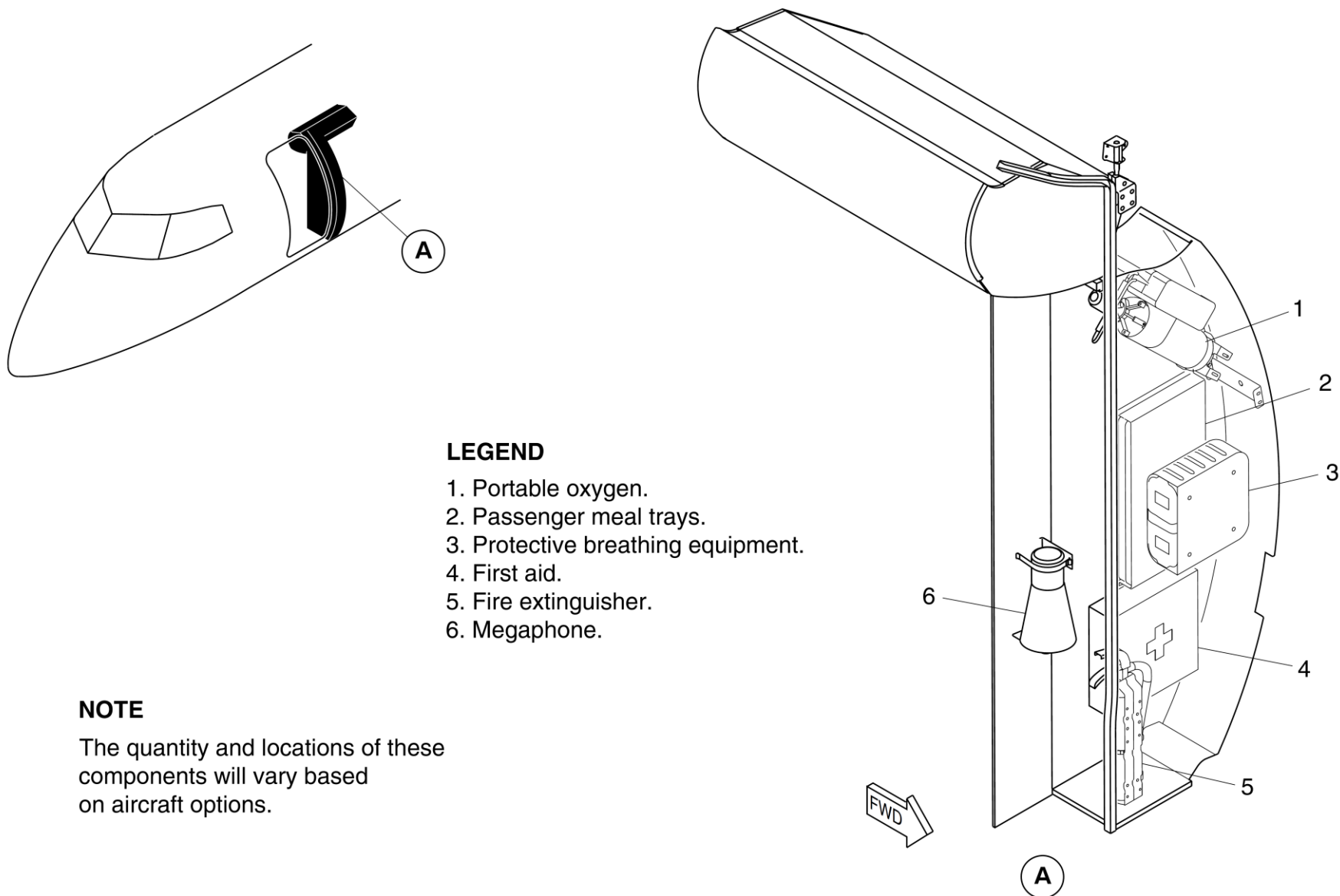


Figure 35-12 Emergency Equipment - General Layout



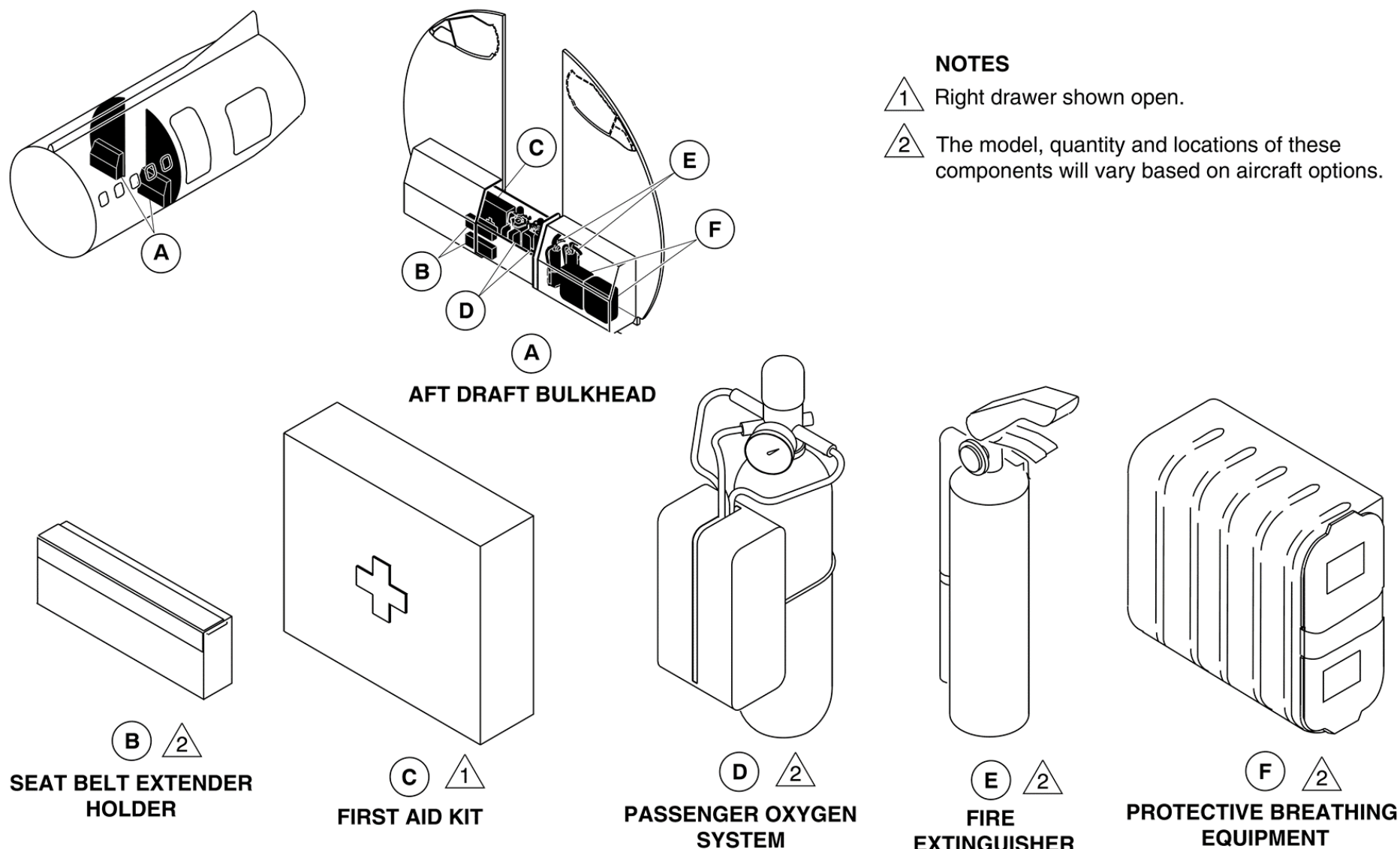


Figure 35-14 Emergency Equipment – Aft Draft Bulkhead

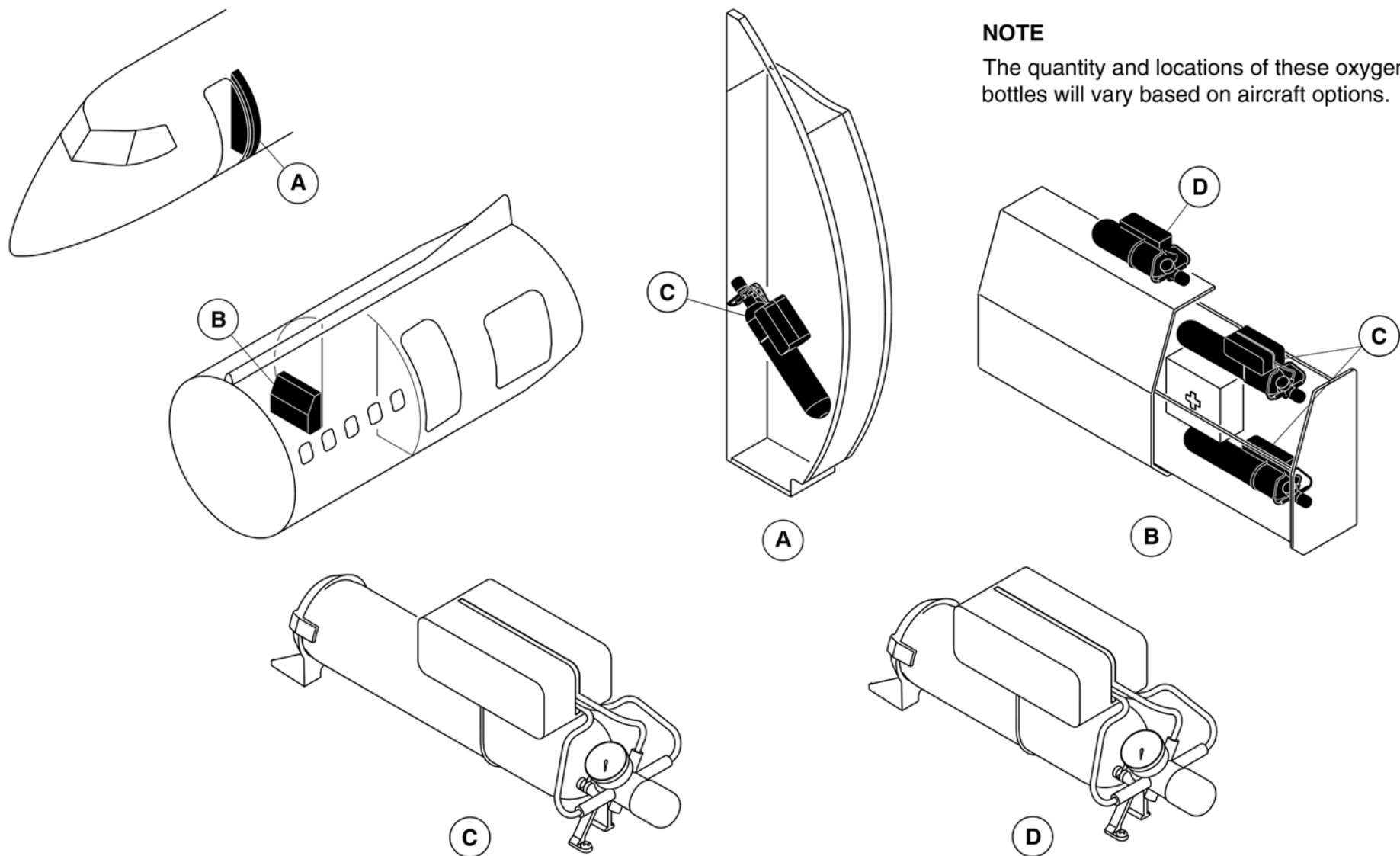


Figure 35-15 Passenger Emergency Oxygen System

Protective Breathing Equipment

Introduction

The Protective Breathing Equipment (PBE) protects the crew member from the effects of smoke, carbon dioxide, harmful gases and oxygen deficiency.

General Description

[Refer Figure – Emergency Equipment – Flight Compartment](#)

[Refer Figure – Emergency Equipment – Frwd Draft Bulkhead](#)

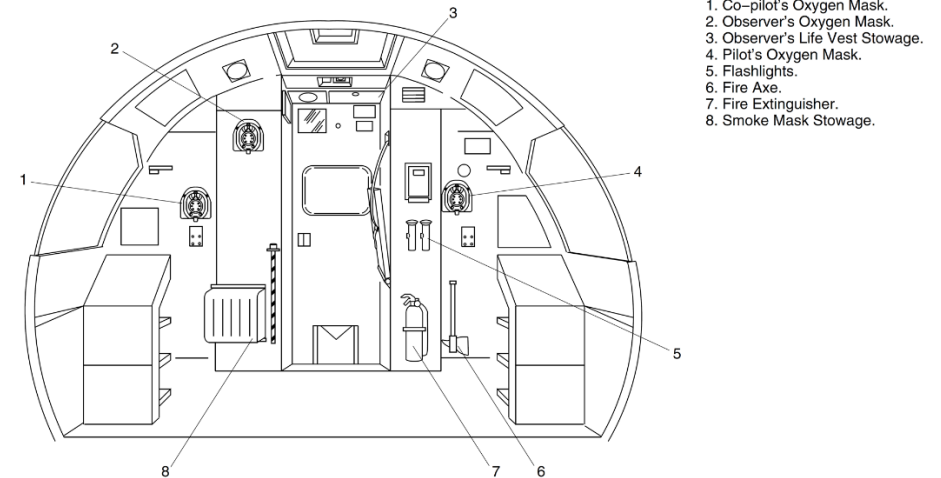
[Refer Figure – Emergency Equipment – Aft Draft Bulkhead](#)

[Refer Figure – Emergency Equipment – General Layout](#)

The PBE is a self contained, portable, disposable breathing device. The system will supply oxygen for approximately 15 minutes during fire, smoke or fume emergencies.

The PBE has a 10 year useful service life. The PBE units are located throughout the aircraft near the fire extinguishers. They supply the crew members with a portable oxygen supply and eye protection while fighting fires in a smoky environment.

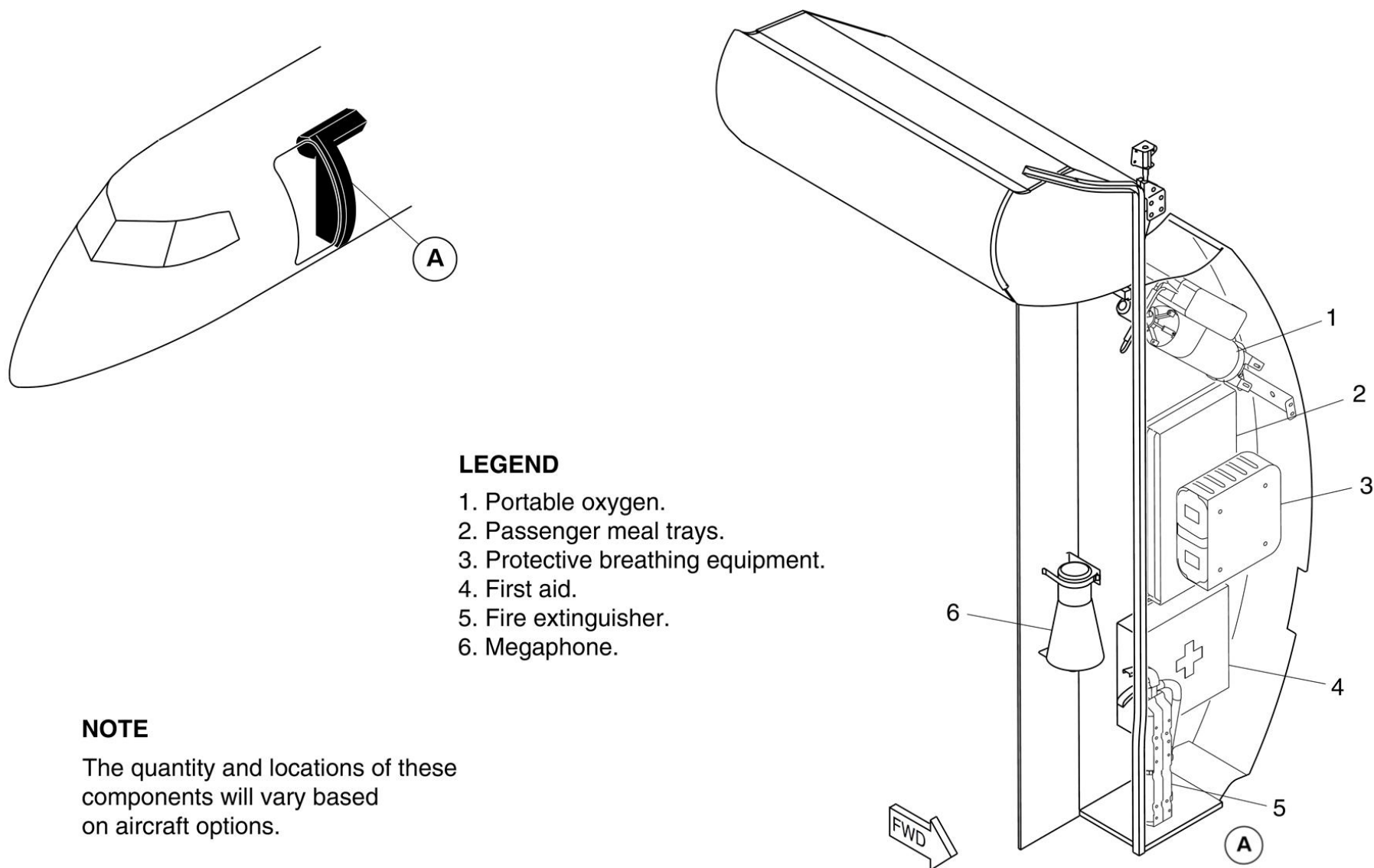
The locations of the PBE units are clearly labelled and the units are easily accessible to the crew.



LEGEND

1. Co-pilot's Oxygen Mask.
2. Observer's Oxygen Mask.
3. Observer's Life Vest Stowage.
4. Pilot's Oxygen Mask.
5. Flashlights.
6. Fire Axe.
7. Fire Extinguisher.
8. Smoke Mask Stowage.

Figure 35-16 Emergency Equipment – Flight Compartment



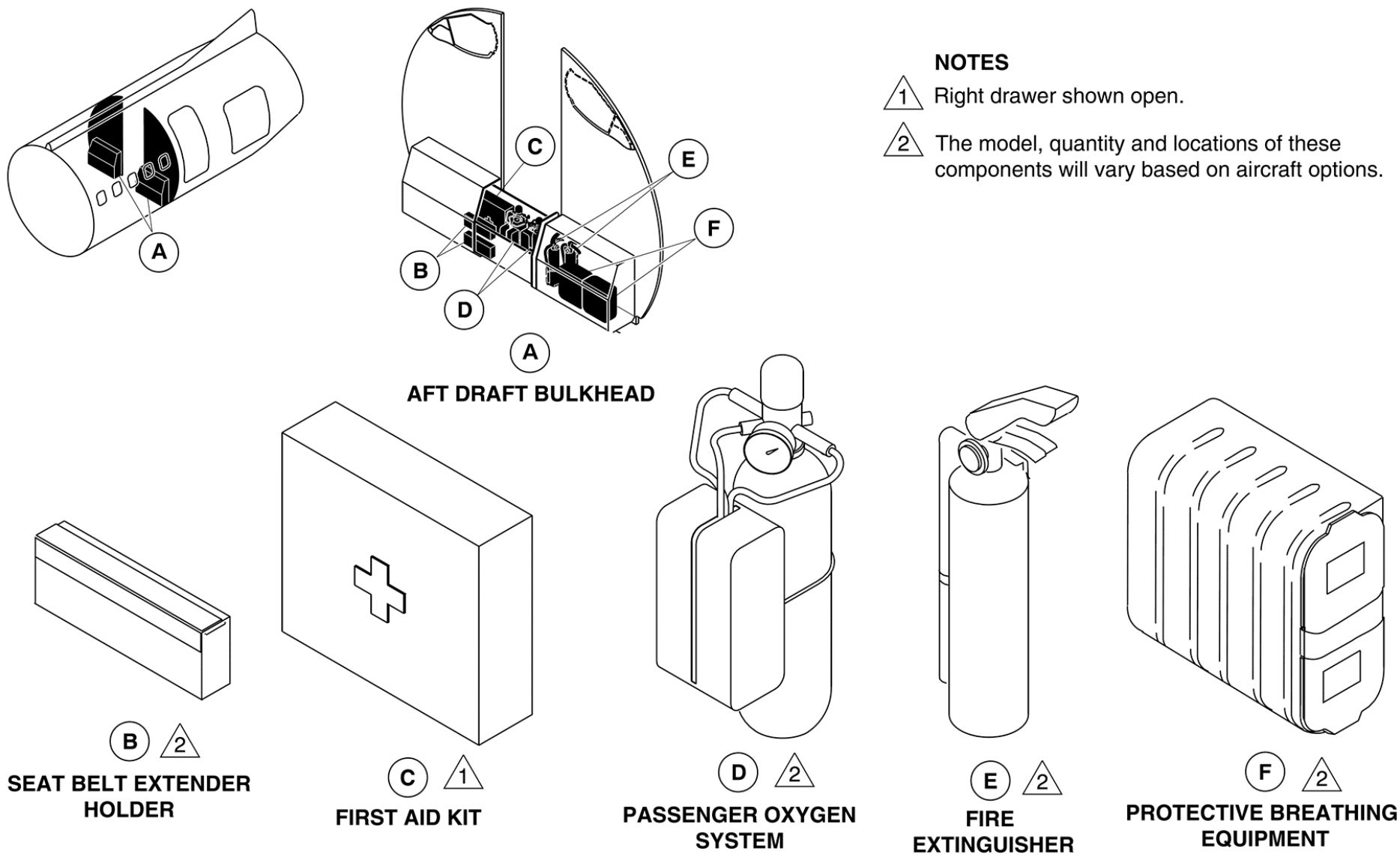


Figure 35-18 Emergency Equipment – Aft Draft Bulkhead

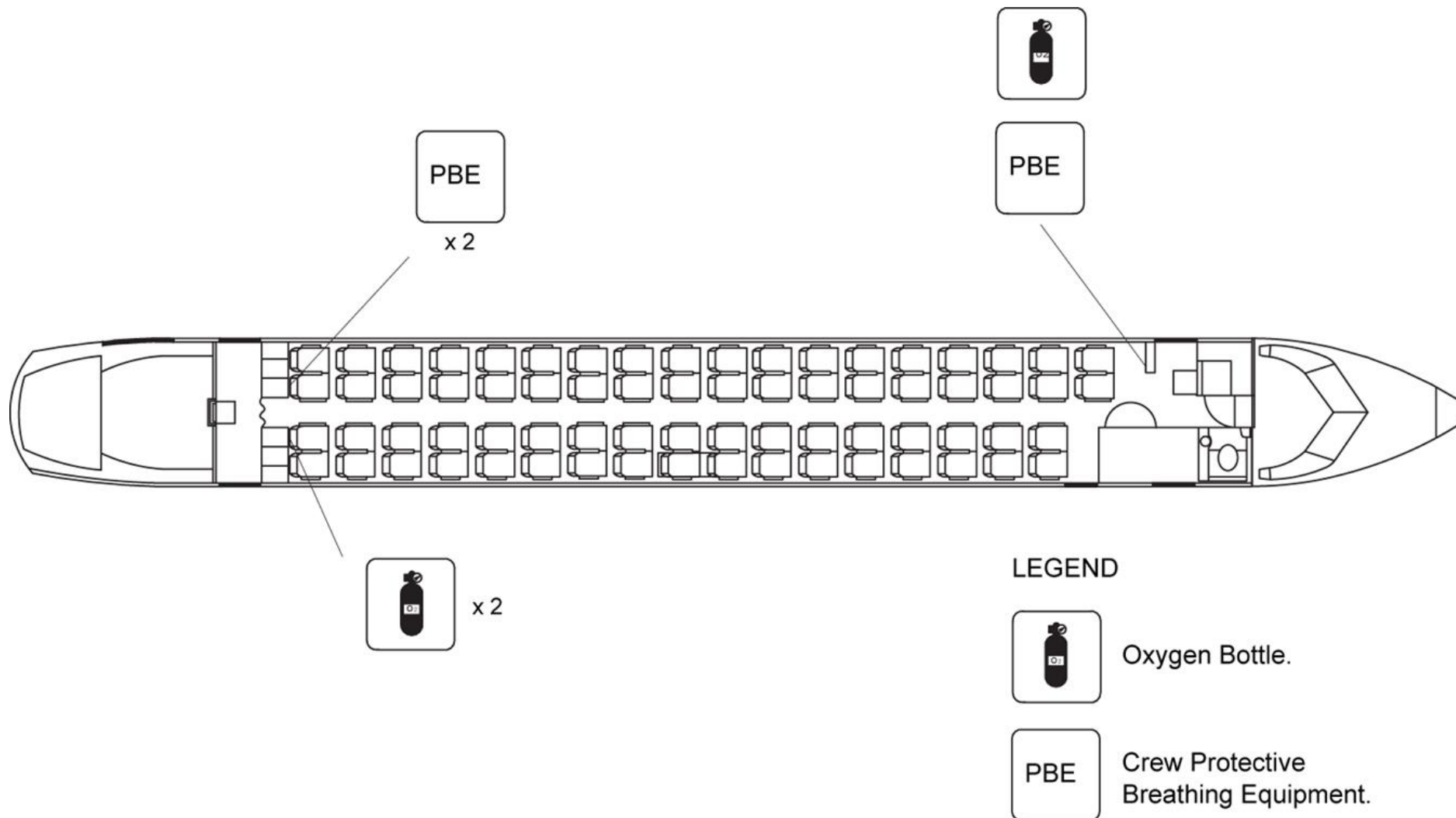


Figure 35-19 Emergency Equipment – General Layout

Detailed Description

[Refer Figure – Protective Breathing Equipment](#)

There are four PBE units, one in the flight compartment and three in the passenger compartment. In the flight compartment, the PBE unit is located in a storage container attached to the bulkhead behind the copilot's seat. It is easily accessible to both pilots.

In the passenger compartment, the PBE's are located in storage containers near the fire extinguishers: one in the forward draft bulkhead and two in the aft draft bulkhead.

The PBE unit is a self-contained breathing device. It is stored in a container with an easily removed see-through door. The unit is kept in a vacuum sealed pouch for protection from moisture and contamination. The unit is ready for use as long as a pull strip on the sealed pouch is intact and there is no visual indication of loss of vacuum. Pulling the strip breaks the vacuum seal and lets the wearer use the protective breathing equipment.

The PBE unit is a demand system. The exhaled air from the wearer passes through a canister of potassium dioxide. The water vapor and carbon dioxide from the exhaled air is converted to oxygen which can be used by the wearer.

The rate of conversion to oxygen is based on the work rate demanded by the wearer. The canister is sized to supply a minimum of 15 minutes of oxygen at pressure altitudes of up to 8000 feet.

Protective Breathing Equipment - Component Description

Hood

On aircraft installed with BE Aerospace 119003 SERIES PBE's, the hood encloses the head of the wearer and seals around the neck with a thin elastic membrane and the large internal volume of the hood accommodates glasses and long hair. The enclosed volume of the hood represents approximately eight liters available re breathing volume. Inhalation is accomplished directly from the interior of the hood through the inhalation valve located on the oronasal mask. The complete device is secured to the head to minimize restrictions of mobility.

A neck shield extends downward from the back of the hood to protect the collar and upper shoulder of the user from direct flame contact. A speaking diaphragm is installed in the oronasal mask cone to enhance the communication.

A relief valve located at the back of the hood prevents excessive pressure from building up inside the hood, and provides emergency pressure relief if cabin pressure is suddenly reduced. The chlorate candle mounted on the bottom of the potassium superoxide (KO_2) canister discharges through the canister and into the hood. When the neck seal is properly positioned, the oxygen generated by the chlorate candle will inflate the hood, provide adequate initial breathing volume, and accomplish some purging of the hood with excess gas generation.

Oxygen System

On aircraft installed with BE Aerospace 119003 SERIES PBE's, chemically supplied oxygen with a capacity of approximately 50 liters on demand with a duration of 15 min or longer when used in accordance with specified work rates, depending on the size of the user. Chlorate Candle.

On aircraft installed with BE Aerospace 119003 SERIES PBE's, the chlorate candle is designed to deliver a fixed volume of approximately eight liters of

oxygen in less than 20 seconds.

The chlorate candle is contained in a small stainless steel cylinder mounted on the base of the potassium superoxide (KO_2) canister. Discharge from it is directed into the interior of the potassium superoxide (KO_2) canister.

When the lanyard attached to the right-hand adjustment strap is pulled, a spring-loaded plunger strikes a small percussion primer to initiate the chlorate candle.

System Operation

On aircraft installed with BE Aerospace 119003 SERIES PBE's, during the donning sequence, the chlorate candle is actuated when the adjustment straps are pulled to secure the oronasal mask cone against the face.

The lanyard leading to the pull pin on the chlorate candle is attached to the right-hand adjustment strap. When the wearer begins to breathe normally through the oronasal mask cone, the exhaled breathe is routed through the potassium superoxide (KO_2) canister.

The potassium superoxide (KO_2) canister removes water vapor and carbon dioxide and adds oxygen before the gas is returned to the interior of the hood for inhalation. This breathing cycle will continue until the potassium superoxide (KO_2) canister is exhausted.

Exhaustion of the canister results in a loss of active oxygen generation capability, coupled with a rapid increase in internal temperature and release of moisture from the potassium superoxide (KO_2) canister.

The loss of oxygen generating capability results in a gradual reduction of the interior volume of the hood. As the hood collapses around the wearer's head, full inhalation becomes increasingly difficult, indicating that the hood should be removed. The rapid rise in temperature inside the hood reinforces this indication.

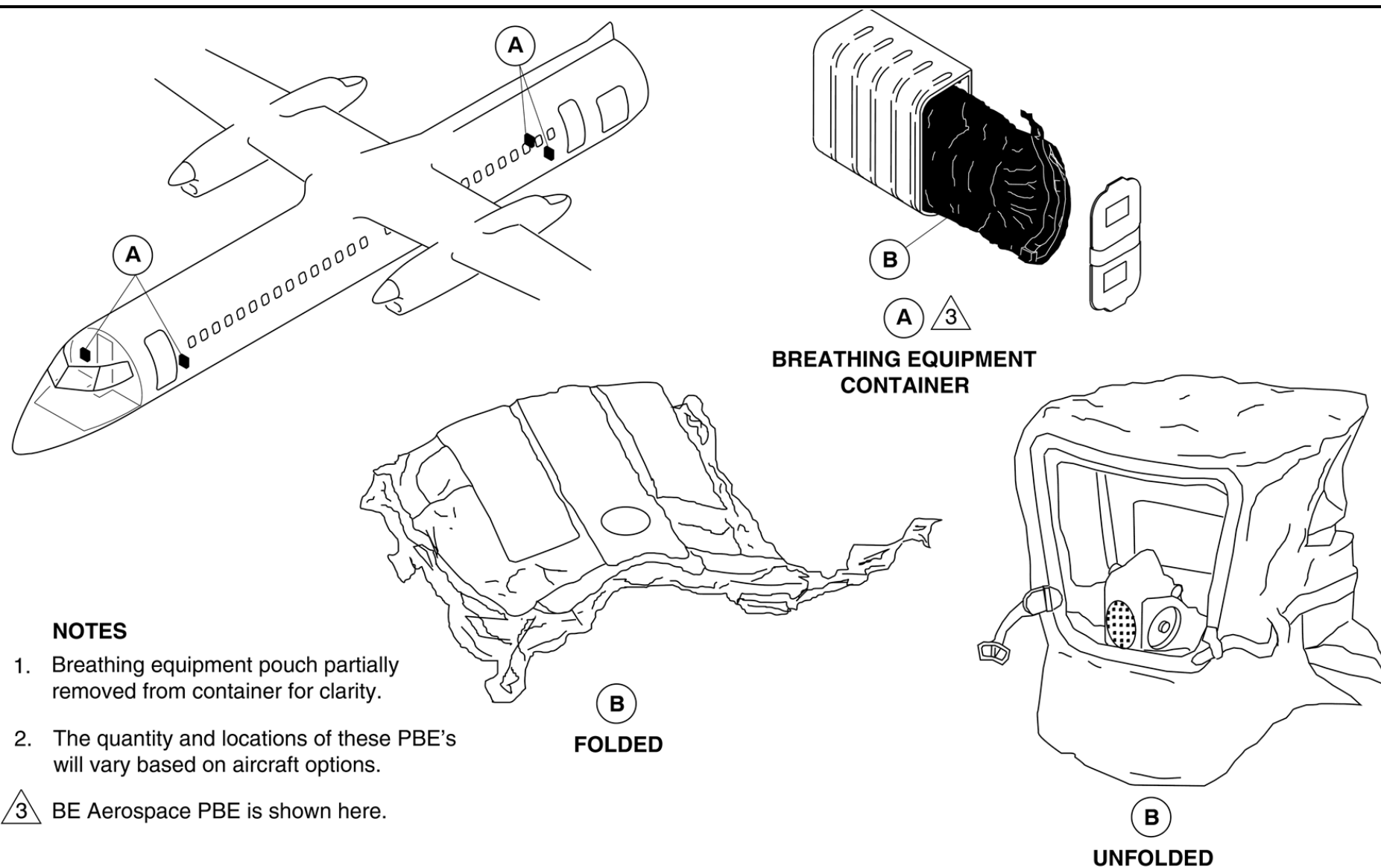


Figure 35-20 Protective Breathing Equipment